

Home Lab Documentation

Ubuntu Server & Windows Client – System Integration Basics

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Role Target: Ausbildung – Fachinformatiker für Systemintegration

Environment: VirtualBox Home Lab

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1. Objective

The goal of this home lab is to gain practical experience in basic system administration and networking by setting up and managing a Linux server in a virtualized environment.

The lab focuses on:

- Installing and configuring Ubuntu Server
- Managing users and permissions
- Installing and controlling system services
- Understanding basic networking concepts
- Preparing the system for remote administration via SSH

2. Lab Environment Overview

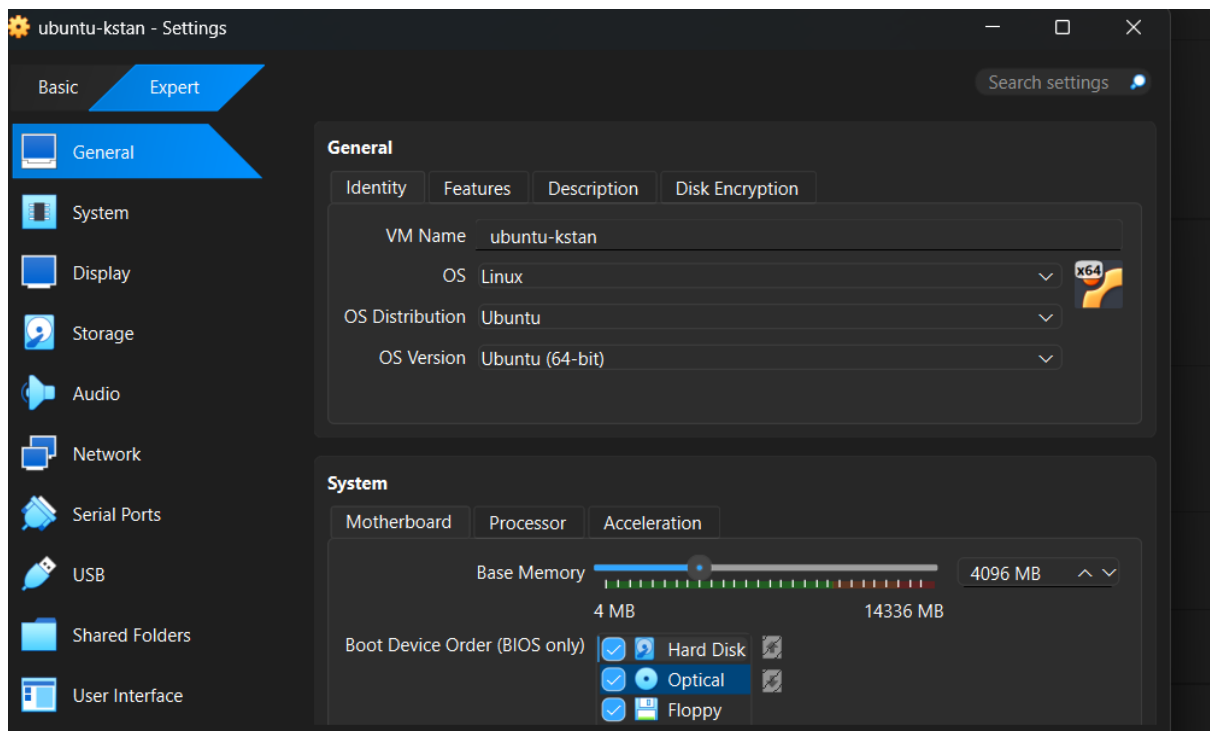
Host System

- Host OS: Windows 11 Pro
- Virtualization Software: VirtualBox

Virtual Machines

- **Ubuntu Server 24.04 LTS**
 - Role: Server
 - Network mode: NAT (initial setup)
 - Purpose: Linux administration and service management
- **Windows Client**
 - Role: Client system
 - Purpose: Remote access and connectivity testing

VirtualBox VM Overview



3. Ubuntu Server Installation

Ubuntu Server 24.04 LTS was installed using the official ISO image. The installation was performed in text-based (headless) mode.

Key installation settings:

- Language: English
- User created during installation: kstan
- OpenSSH Server: not initially installed
- Disk: Automatically partitioned
- Network: DHCP via NAT

After installation, the system booted successfully to a login prompt.

Ubuntu Server Login Prompt

```
Ubuntu 24.04.3 LTS ubuntu-kstan tty1
ubuntu-kstan login: kstan
Password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-90-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Jan 17 11:57:56 AM UTC 2026

System load:          0.15
Usage of /:            10.9% of 24.44GB
Memory usage:         6%
Swap usage:           0%
Processes:            109
Users logged in:      0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe3f:e308

Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

kstan@ubuntu-kstan:~$ sudo apt update && sudo apt upgrade -y_
```

4. System Update and Basic Tools

After the first login, the system was updated and basic administrative tools were installed.

Commands executed:

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install net-tools curl wget htop -y
```

Purpose:

- Ensure system security and stability
- Install common diagnostic and monitoring tools

System Update / Package Installation

```
Swap usage:          0%
Processes:           109
Users logged in:     0
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fe9f:e308

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

kstan@ubuntu-kstan:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for kstan:
Hit:1 http://de.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://de.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://de.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,694 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
Get:7 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:8 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.4 kB]
Get:10 http://de.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.7 kB]
Get:12 http://de.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1,514 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Get:14 http://de.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:15 http://de.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [31.5 kB]
Get:16 http://de.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:17 http://de.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,284 B]
Get:18 http://de.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:19 http://de.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:20 http://de.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Fetched 4,302 kB in 1s (2,886 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
2 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  kpartx multipath-tools
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
kstan@ubuntu-kstan:~$

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

kstan@ubuntu-kstan:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for kstan:
Hit:1 http://de.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:3 http://de.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:4 http://de.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://de.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1,694 kB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.6 kB]
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Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [19.7 kB]
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Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
The following upgrades have been deferred due to phasing:
  kpartx multipath-tools
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
kstan@ubuntu-kstan:~$ sudo apt install net-tools curl wget httpd -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
net-tools is already the newest version (2.10-0.1ubuntu4.4).
curl is already the newest version (8.5.0-2ubuntu10.5).
wget is already the newest version (1.21.4-1ubuntu4.1).
httpd is already the newest version (3.3.0-4build1).
0 upgraded, 0 newly installed, 0 to remove and 2 not upgraded.
kstan@ubuntu-kstan:~$
```

5. Network Verification

The network configuration was verified to ensure proper connectivity.

Commands used:

ip a

ip route

ping 8.8.8.8

ping google.com

Results:

- Network interface enp0s3 was active
- IPv4 address assigned via DHCP (NAT network)
- Internet connectivity and DNS resolution confirmed

ip a Output

```
64 bytes from lcnuca-ai-in-f14.1e100.net (172.217.19.174): icmp_seq=143 ttl=255 time=16.2 ms
64 bytes from lcnuca-ai-in-f14.1e100.net (172.217.19.174): icmp_seq=144 ttl=255 time=16.6 ms
64 bytes from lcnuca-ai-in-f14.1e100.net (172.217.19.174): icmp_seq=145 ttl=255 time=13.4 ms
64 bytes from lcnuca-ai-in-f14.1e100.net (172.217.19.174): icmp_seq=146 ttl=255 time=13.6 ms
64 bytes from lcnuca-ai-in-f14.1e100.net (172.217.19.174): icmp_seq=147 ttl=255 time=16.1 ms
^C
--- google.com ping statistics ---
147 packets transmitted, 147 received, 0% packet loss, time 157095ms
rtt min/avg/max/mdev = 9.268/19.784/242.281/21.277 ms
kstan@ubuntu-kstan:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:3f:e3:88 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86012sec preferred_lft 86012sec
    inet6 fd17:625c:f037:2:a00:27ff:fe3f:e388/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86186sec preferred_lft 14186sec
    inet6 fe80::a00:27ff:fe3f:e388/64 scope link
        valid_lft forever preferred_lft forever
kstan@ubuntu-kstan:~$
```

6. SSH Installation and Service Management

Initially, the SSH service was not available. OpenSSH Server was installed manually.

Installation:

sudo apt install openssh-server -y

Service configuration:

sudo systemctl disable ssh.socket

sudo systemctl stop ssh.socket

sudo systemctl enable ssh

sudo systemctl start ssh

This ensured that the SSH service runs continuously instead of being socket-activated.

Verification:

systemctl status ssh

Result:

- SSH service installed
- Service configured and controlled using systemd

```
kstan@ubuntu-kstan:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-01-17 11:57:20 UTC; 59min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Process: 735 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 752 (sshd)
      Tasks: 1 (limit: 4603)
     Memory: 2.2M (peak: 2.4M)
        CPU: 46ms
    CGroup: /system.slice/ssh.service
            └─752 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Warning: some journal files were not opened due to insufficient permissions.
kstan@ubuntu-kstan:~$
```

7. SSH Connectivity Issue (Learning Outcome)

An SSH connection attempt from the Windows host to the Ubuntu server failed with a timeout error while using NAT networking.

Error:

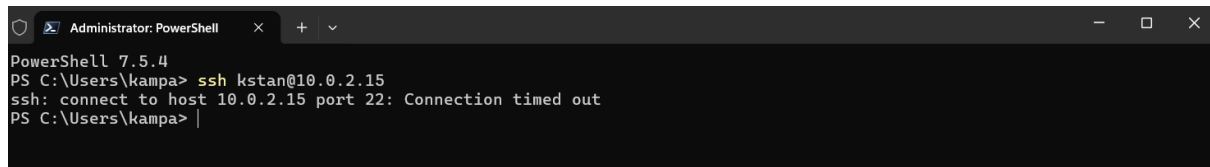
ssh: connect to host 10.0.2.15 port 22: Connection timed out

Analysis:

- SSH service was running
- NAT networking does not allow inbound connections by default
- This demonstrated the practical difference between NAT and Bridged networking

This issue was identified as a **network topology limitation**, not a system failure.

SSH Timeout Error (Windows PowerShell)



```
PowerShell 7.5.4
PS C:\Users\kampa> ssh kstan@10.0.2.15
ssh: connect to host 10.0.2.15 port 22: Connection timed out
PS C:\Users\kampa>
```

8. User Management and Privilege Control

A second user was created to practice Linux user and permission management.

User creation:

```
sudo adduser testuser
```

Verification:

```
id testuser
```

```
groups testuser
```

Granting sudo rights:

sudo usermod -aG sudo testuser

Privilege test:

su - testuser

sudo whoami

Result:

- testuser successfully escalated privileges using sudo
- Proper separation between users and administrative rights established

User and Group Verification

```
kstan@ubuntu-kstan:~$ groups anna
anna : anna users
kstan@ubuntu-kstan:~$ id testuser
uid=1001(testuser) gid=1001(testuser) groups=1001(testuser),27(sudo),100(users)
kstan@ubuntu-kstan:~$ groups testuser
testuser : testuser sudo users
kstan@ubuntu-kstan:~$
```

9. File Permissions and Access Control

A shared directory was created to practice permissions and group-based access.

Commands used:

sudo mkdir /srv/shared

sudo chown root:sudo /srv/shared

sudo chmod 770 /srv/shared

Test as testuser:

cd /srv/shared

touch testfile

Result:

- Access granted only to root and sudo group members
- Permission model successfully implemented

Shared Directory Permissions

```
kstan@ubuntu-kstan:~$ su -testuser
su: invalid option -- 't'
Try 'su --help' for more information.
kstan@ubuntu-kstan:~$ su - testuser
Password:
su: Authentication failure
kstan@ubuntu-kstan:~$ su - testuser
Password:
testuser@ubuntu-kstan:~$ groups
testuser sudo users
testuser@ubuntu-kstan:~$ cd /srv/shared
testuser@ubuntu-kstan:/srv/shared$ touch testfile
testuser@ubuntu-kstan:/srv/shared$ ls -ld /srv/shared
drwxr-x-- 2 root sudo 4096 Jan 17 09:24 /srv/shared
testuser@ubuntu-kstan:/srv/shared$ touch /srv/shared/testfile
testuser@ubuntu-kstan:/srv/shared$ cd /srv/shared
testuser@ubuntu-kstan:/srv/shared$ touch testfile
testuser@ubuntu-kstan:/srv/shared$
```

10. Skills Demonstrated

Through this home lab, the following skills were demonstrated:

- Linux server installation and configuration
- Package management with APT
- Network verification and troubleshooting
- System service management using systemd
- SSH installation and access concepts
- User and group administration
- Linux file permissions and access control
- Understanding of NAT networking limitations

11. Next Steps

Planned next steps include:

- Switching the VM to Bridged networking
- Establishing successful SSH access from Windows
- Firewall configuration using UFW
- Installing and exposing a basic service (Apache)
- Continued documentation and system hardening